

Dayton Shipment Analytics & Predictive Modeling Project

An end-to-end portfolio project demonstrating data generation, data quality engineering, microeconomic analysis, and machine-learning forecasting in a fictional Pokemon-universe trucking operation.

Project Overview

This project simulates an end-to-end analytics workflow for **Dayton**, a fictional trucking company operating across major Pokemon-region commercial hubs. The project follows the complete data lifecycle: first creating a realistic operational database, then establishing trust through an automated cleaning pipeline, and finally using the cleaned historical data to predict future business performance.

Section	Purpose	Core Skills
1 - Build the Data	Create a realistic but intentionally imperfect shipment database.	Data modeling, logistics, microeconomics
2 - Trust the Data	Clean, validate, repair, classify, and audit shipment records.	Python, pandas, ETL, QA, business rules
3 - Use the Data	Forecast future revenue and profit from trusted historical data.	Feature engineering, Gradient Boosting, forecasting

Section 1 - Building the Shipment Database

Objective

The first stage created a fictional operational shipment database representing Dayton's trucking activity across major cities in the Pokemon world. Approximately **1,000 shipment records** were generated. The database was intentionally designed to resemble a real operational export containing both valid data and controlled data-quality problems.

Transportation Network

- Kanto - Saffron City
- Johto - Goldenrod City
- Hoenn - Lilycove City and Slateport City
- Sinnoh - Jubilife City
- Unova - Castelia City
- Kalos - Lumiose City
- Alola - Hau'oli City
- Galar - Wyndon
- Paldea - Mesagoza

A standardized lane-distance matrix was developed for every possible origin-destination combination. With 10 hubs, the network contains **45 unique city pairs and 90 directional lanes**. Mileage is symmetric, so the same standard distance applies in both directions. The matrix became the project's source of truth for shipment-mileage validation.

Shipment Commodities

- Poke Balls - including Poke Ball, Great Ball, Ultra Ball, Net Ball, and Quick Ball.
- Healing Items - including Potion, Super Potion, Hyper Potion, Max Potion, and Full Restore.
- Status Healers - including Antidote, Burn Heal, Ice Heal, Awakening, Paralyz Heal, and Full Heal.
- Escape & Miscellaneous Items - including Escape Rope, Repel, Super Repel, and Max Repel.

Microeconomic Structure

The shipment report was also designed as an economic dataset. Each shipment contains operating costs, revenue, and profitability measurements, allowing core microeconomic relationships to be applied directly to logistics activity.

$$\text{TVC} = \text{Fuel Cost} + \text{Driver Cost} + \text{Toll Cost} + \text{Handling Cost}$$

$$\text{TC} = \text{TFC} + \text{TVC}$$

$$\text{TR} = \text{Quantity} \times \text{Revenue Per Unit}$$

$$\text{Profit} = \text{TR} - \text{TC}$$

$$\text{AVC} = \text{TVC} / \text{Quantity}$$

$$\text{ATC} = \text{TC} / \text{Quantity}$$

Intentionally Introduced Data Problems

- Duplicate and conflicting shipment records.
- Missing cities, regions, and completely missing origin/destination geography.
- Inconsistent customer names such as ACME, Acme Inc., ACME INC, and ACME Incorporated.
- Dates stored in multiple formats.
- Negative shipment quantities.
- Missing mileage and missing cost measurements.
- Missing revenue and implausible revenue rates.
- Invalid shipment statuses.
- Incorrect variable cost, total cost, and profit calculations.
- Negative or otherwise inconsistent economic values.

Section 1 therefore establishes the project's central data-quality challenge: which records can safely be repaired automatically, and which records should not be trusted without human review?

Section 2 - Python Data Quality & Cleaning Pipeline

Objective

The second stage develops a Python-based ETL and data-quality pipeline that converts the intentionally messy shipment database into a trusted analytical dataset. Rather than simply deleting nulls and duplicates, the pipeline applies explicit business rules and preserves an audit trail.

Record Classification

- **CLEAN** - record required no substantive correction.
- **AUTO_CORRECTED** - record could be repaired safely using deterministic rules or robust benchmarks.
- **NEEDS_REVIEW** - ambiguity remains and a human decision is required.
- **REJECTED** - the record contains an unsafe or uninterpretable business value.
- **REMOVED** - the record is excluded from the reporting population by business rule.

1. Validate Geography

A shipment is removed when either the origin has both Region and City blank, or the destination has both Region and City blank. Partial geography is repaired only when the missing value can be inferred safely. For example, Wyndon uniquely maps to Galar. Ambiguous cases such as Hoenn with a missing city are sent to NEEDS_REVIEW because Hoenn contains both Lilycove City and Slateport City in the modeled network.

2. Apply Shipment Status Rules

Only the following operational statuses remain in the analytical population:

- Dispatched
- Delivered
- DONE
- In Transit
- Delayed

All other statuses are removed.

3. Validate Shipment Quantity

Missing, zero, or negative quantities are rejected rather than automatically converted. A negative value could represent a reversal, return, correction, or source-system error, so the pipeline does not guess.

4. Restore Standard Mileage

Resolved origin and destination cities are matched to the lane-distance matrix created in Section 1. The matrix replaces missing or inconsistent mileage and ensures that each lane uses a standardized distance.

5. Repair Missing Economic Measurements

Missing base-cost components are estimated from robust statistics calculated from valid records. The pipeline uses median Fuel Cost per Mile, Driver Cost per Mile, Toll Cost per Mile, Handling Cost per Unit, and Fixed Cost Allocation. Medians are used because they are less sensitive to extreme or corrupted values.

6. Reconstruct the Economics

Derived economic fields are not repaired independently. Instead, the pipeline rebuilds them from trusted base measurements. This is a critical design choice because it restores internal mathematical consistency.

Fuel + Driver + Tolls + Handling -> Variable Cost

Variable Cost + Fixed Cost -> Total Cost

Quantity x Revenue Per Unit -> Total Revenue

Total Revenue - Total Cost -> Profit

Variable Cost / Quantity -> AVC

Total Cost / Quantity -> ATC

7. Standardize Customers

Known customer-name variants are normalized to a single canonical value. For example, ACME, Acme Inc., acme inc, and ACME Incorporated are standardized to **ACME INC.**

8. Process Duplicates

Exact duplicates are safely deduplicated by retaining one copy. Conflicting duplicates are not resolved arbitrarily; all remaining records associated with the conflicting Shipment_ID/Order_ID are quarantined as NEEDS_REVIEW.

Pipeline Outputs

- Dayton_Clean_Shipment_Report.csv
- Dayton_Auto_Corrected_Records.csv
- Dayton_Needs_Review.csv
- Dayton_Rejected_Records.csv
- Dayton_Removed_Records.csv
- Dayton_Data_Quality_Audit.csv
- Dayton_Data_Quality_Summary.csv
- Dayton_Anomaly_Validation_Crosswalk.csv

Section 2 produces both a clean analytical dataset and a transparent record of what happened to every shipment, demonstrating data governance rather than opaque data manipulation.

Section 3 - Machine Learning & Future Forecasting

Objective

The final stage uses only the cleaned shipment report to determine whether Dayton's historical shipment economics can support predictions of future business performance. This separation is intentional: machine learning begins only after the data has passed through the quality pipeline.

Weekly Time-Series Transformation

Individual shipments are aggregated into weekly business-performance observations. The modeling dataset tracks Weekly Revenue, Weekly Profit, Weekly Cost, Shipment Count, Total Quantity, and Average Distance.

Model Selection

Gradient Boosting Regression was selected for the forecasting stage because the project uses a small-to-medium structured dataset with potentially nonlinear relationships between historical performance, trend, and seasonality.

Model	Prediction Target
Model 1	Weekly Total Revenue
Model 2	Weekly Profit

Feature Engineering

The forecasting models use historical behavior rather than relying on the calendar date alone. Features include:

- Sequential trend index.
- Week of year, month, and quarter.
- Cyclical sine and cosine seasonality terms.
- 1-week, 2-week, 4-week, and 13-week lagged target values.
- 4-week and 13-week rolling averages.
- 4-week rolling standard deviation to represent recent volatility.

Time-Aware Model Validation

Historical observations are split chronologically rather than randomly. Earlier weeks form the training set and later unseen weeks form the test set. This prevents future observations from leaking into the past and more closely reflects a real forecasting environment.

Model Evaluation

- MAE - Mean Absolute Error.
- RMSE - Root Mean Squared Error.
- R-squared - proportion of target variation explained by the model.
- MAPE - Mean Absolute Percentage Error, where applicable.
- Feature importance - relative contribution of engineered predictors.

52-Week Forecast

After evaluation, each model generates a recursive forecast for the next 52 weeks. Predicted values from one future week become part of the lagged history used to forecast later weeks. Revenue is constrained to remain non-negative, while profit is permitted to become negative if the model predicts an unprofitable period.

Forecast Reporting

- Weekly revenue and profit predictions for the next 52 weeks.
- Monthly forecast summaries.
- Next-year forecast revenue.
- Next-year forecast profit.
- Implied forecast cost.
- Forecast profit margin.
- Actual-vs-predicted validation graphics.
- Feature-importance graphics.
- Historical plus 52-week forecast visualizations.

Section 3 converts trusted historical operational data into forward-looking business intelligence, moving the project beyond descriptive reporting into predictive analytics.

Complete Project Architecture

SECTION 1 - BUILD THE DATA	1,000 fictional shipments; transportation network; standardized lane matrix; operational measurements; microeconomic measurements; intentional data problems.
SECTION 2 - TRUST THE DATA	Python data-quality pipeline; validation; business rules; economic reconstruction; exception management; customer standardization; audit trail.
SECTION 3 - USE THE DATA	Weekly aggregation; feature engineering; Gradient Boosting; chronological validation; revenue forecasting; profit forecasting; 52-week prediction.

Overall analytical story:

Raw operational data -> data governance and quality -> economic analysis -> predictive analytics -> future business decision support.

The project demonstrates the full lifecycle of a business analytics problem rather than presenting disconnected scripts. It shows how a data analyst can create and understand a dataset, establish whether it is trustworthy, apply operational and economic business rules, and finally transform that trusted information into a predictive model designed to support future planning.